

PROTOCOL TO COLLECT CAUSAL CHAIN DATA ON AGRICULTURE ADAPTATION EFFECTIVENESS AND RESILIENCE



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Titles in this series aim to disseminate interim climate change, agriculture and food security research and practices and stimulate feedback from the scientific community.

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ABOUT AICCRA



AICCRA
Accelerating Impacts of CGIAR
Climate Research for Africa



Accelerating Impacts of CGIAR Climate Research for Africa (AICCRA) is a project that helps deliver a climate-smart African future driven by science and innovation in agriculture. It is led by the Alliance of Bioversity International and CIAT and supported by a grant from the International Development Association (IDA) of the World Bank. Explore our work at aiccra.cgiar.org



ABSTRACT

This working paper introduces a conceptual model and operational protocol for assessing adaptation effectiveness and resilience in agricultural systems through explicit causal chains linking climate shocks, adaptive actions, mediating processes, and outcomes. Building on a rich literature on resilience, vulnerability, and adaptation measurement, it addresses persistent fragmentation in the evidence base, where different parts of the adaptation process are often analyzed in isolation. The paper presents a structured model and two complementary tools—a household survey module and a participatory focus group guide—that enable the collection of relational, event-anchored data across shocks, actions, and effects. Applied within the *Accelerating Impacts of CGIAR Climate Research for Africa* (AICCRA) project, the protocol demonstrates how these tools can operationalize complex impact pathways without imposing new data burdens. The novelty of this effort lies in providing a coherent causal structure that connects actions, mechanisms, and outcomes over time; integrating monitoring, evaluation, and impact assessment efforts within a single logic; and bridging empirical and experiential evidence to generate a dynamic, cumulative understanding of how adaptation produces resilience across contexts and timescales.

Keywords

Agriculture; climate change; adaptation; resilience; effectiveness; data collection

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Contents

Abstract	1
CAUSAL CHAINS FOR ADAPTATION EFFECTIVENESS AND RESILIENCE	3
Fragmented evidence on agricultural climate adaptation effectiveness and resilience	3
Causal chains for adaptation effectiveness and resilience	6
Causal chain components.....	7
Scale of application	9
Alignment with current approaches	9
TOOLS FOR CAUSAL CHAIN DATA COLLECTION	10
Survey tool	10
Core questions	11
Sampling	15
Design notes.....	15
Implementation notes	16
Interpreting resilience signals	17
Caveats.....	18
Focus group discussions (FGDs).....	20
Core questions	20
Sampling	22
FGD Implementation	22
Analysis and interpretation	24
FGD Limitations.....	25
USE CASE: AICCRA	27
Survey implementation.....	28
FGD implementation.....	29
Conclusion.....	31
Appendix	33
Appendix 1. AICCRA Survey: Assessing linkages between capacities, context, and resilience outcomes.....	33
References	36



CAUSAL CHAINS FOR ADAPTATION EFFECTIVENESS AND RESILIENCE

Fragmented evidence on agricultural climate adaptation effectiveness and resilience

The science of agricultural adaptation is rich (Barbieri et al. 2024; Challinor et al. 2014; Dolan et al. 2001; IPCC 2022; Obossou et al. 2025; Rosenstock et al. 2024). Over the past decades, researchers and development practitioners have built an array of frameworks and metrics to understand how farm systems respond to changes and variabilities in climate. These efforts capture many facets of the adaptation process—exposure and sensitivity to stressors, adaptive capacities, resource use, behavioral practices, and outcomes such as system stability or vulnerability reduction (van der Lee et al. 2022). This diversity reflects both intellectual vitality and the reality of agricultural systems, which are dynamic, context-specific, and shaped by multiple interacting processes. Yet, this richness has come at a cost. While we have learned much about *pieces* of resilience, we lack a coherent view of how these pieces connect into a whole. Specifically, how climate shocks and stressors interact with capacities and practices to generate enhanced, stable or transformative outcomes.

An analysis of 123 peer-reviewed studies mapped how agricultural resilience has been theorized and identified four dominant lenses—traditional, vulnerability, capacities, and agroecology—each addressing different elements of the adaptation process (van der Lee et al. 2022). The traditional lens – also known as “foundational school of theory on resilience of social-ecological systems” (p.4), focuses on perturbations and system recovery but seldom links these to resources or capacities. The vulnerability lens captures exposure and sensitivity yet overlooks adaptive mechanisms. The capacities lens emphasizes absorptive, adaptive, and transformative abilities but often isolates them from practices or outcomes. The agroecology lens highlights diversity and feedback within systems but weakly connects these to actor-level decisions. All four have looked into outcomes such as stability, transformation, or reduced vulnerability, yet few trace the causal linkages among shocks, resources, practices, and capacities. The authors outline five design choices for better integration—traits and functions, perturbations, resilience type, contributing factors, and outcomes—but no operational roadmap exists for linking these dimensions. Consequently, we know *what* to integrate, but not *how*.

Author authors arrived at a similar conclusion from a different vantage point (Pret et al. 2025). Their review of empirical farm-system resilience assessments distinguishes three dominant study types. The first explores perceptions—how farmers understand risks, impacts, and their own resilience. The second examines attributes, such as reserves, openness, modularity, feedback tightness, and diversity, that shape system behavior. The third measures performance outcomes, including productivity, profitability, and environmental or social well-being. Few studies combine multiple attributes within or across these dimensions,



and even fewer trace how system properties translate into performance outcomes under stress. Most studies capture isolated parts of the adaptation process—one link of the resilience chain, not the chain itself. The authors’ conclusion echoes van der Lee’s conceptual critique: resilience continues to be observed in pieces—through attributes or outcomes—but seldom as the dynamic interplay between them.

Similar patterns emerge in the operational frameworks reviewed (Table 1), which provide an illustrative rather than comprehensive overview of current practice. While these frameworks vary in scope and purpose, most share the same limitations: they seldom articulate a full action–outcome logic, rarely capture temporal dynamics, and often rely on proxy indicators that approximate resilience rather than measure it directly. Together, they demonstrate methodological progress but also reveal how resilience measurement remains largely static and causally incomplete.

Table 1. Operational frameworks unpack resilience snapshots (select examples, not illustrative of all frameworks available).

Framework / approach	Core strength	Action-outcome logic (causal chain)	Dynamic process (temporal focus)	Measurement (proxy/ capacities vs. outcome)
FAO RIMA-II	Quantitative rigor; standardized multidimensional index	● no causal link between adaptation actions and outcomes	● no before–during–after logic	● proxy-based latent variables
FAO SHARP+	Participatory, farmer-led self-assessment	● no causal reasoning or mechanism tracing	● one-off perception snapshot	● subjective scoring, limited validation
UNDP CoBRA / TANGO	Community-based, context-sensitive diagnostic tools	● captures perceived drivers, lacks systematic causality	● no temporal framing of events	● qualitative, non-standardized indicators
“Resilience Dividend” / “Conditional Moments”	Econometric modeling of welfare dynamics	● captures statistical association, not contribution logic	● dynamic but reliant on long-term panel data	● data-intensive, less scalable
Oxfam Multidimensional Resilience Index	Aggregates household outcomes across wellbeing domains	● no causal pathway linking practices to outcomes	● no explicit before–after sequence	● relies on aggregated proxy indicators
UNEP FI Resilience Impact Framework	Connects finance, impact, and resilience outcomes for investors	● uses ToC, weak field-level validation	● not event- or phase-based	● indicator standardization incomplete
One Acre Fund – Resilience Capacity Index	Field-friendly, simple scoring of household coping capacities	● focuses on potential capacities, not contribution	● no tracking of post-shock outcomes	● self-reported proxies dominate
60 Decibels – Resilience Survey	Scalable, perception-based rapid survey	● captures perceived contribution, no triangulation	● snapshot only, lacks temporal framing	● semi-structured perceptions, partial validation

● gap clearly present; ● gap partially addressed



Source: own compilation based on (60decibels n.d.; Abson et al. 2017; Adaptation & Resilience Investors Collaborative 2024; Barrett and Constanas 2014; D’Errico and Smith n.d.; FAO 2015, 2016; Hughes and Bushell 2013)

This fragmentation is a symptom of enduring “habits” the field has learned to live with—definitional plurality, methodological pragmatism, and institutional routines that reward novelty over coherence. The plurality of definitions surrounding adaptation and resilience is understandable, given the context-specificity of adaptation processes; yet it has largely been used to justify new frameworks rather than to reveal complementarities across existing ones. Methodological opportunism reinforces this pattern. Researchers and practitioners build bespoke tools, partly to fit specific contexts but also to signal innovation, an attribute rewarded far more than alignment. Moreover, incentives for less fragmentation remain weak; monitoring, evaluation, and impact assessment remain separated in most institutional architectures and capacities remain disjointed across the two areas; their integration is often conflated with complexity and cost, when in reality it’s about asking the right questions at the right time and scale to understand how adaptation leads to resilience. These are not inevitable conditions of agricultural adaptation science but choices that can be redefined.

One way to address these is through a structural shift in the way we approach adaptation and resilience. This shift is advanced through this protocol and focuses on collecting data in a more *structured and relational* way, favoring integration into a conceptual and analytical causal model: tracing how shocks, adaptation strategies, and resilience outcomes interconnect rather than treating them as independent phenomena. The second is an epistemic shift and concerns how existing data are used. It is not about collecting more, but about using existing data in smarter ways—connecting disparate datasets, surveys, and observations to reconstruct causal pathways and resilience narratives. While this protocol addresses the former, the latter marks an emerging frontier for adaptation science.

In the next section, we introduce the framework for collecting causal-chain data, outlining how it structures the identification and measurement of linkages between shocks, actions, and outcomes. This is followed by two operational tools that translate the framework into practice: a household survey module that systematically quantifies resilience signals and outcomes, and a focus group discussion (FGD) guide that brings a participatory, farmer-centered perspective to uncover underlying mechanisms, contextual factors, and perceived probabilities that structured surveys alone cannot capture. The subsequent section illustrates how these tools were applied and tested within the five-year initiative *Accelerating Impacts of CGIAR Climate Research for Africa (AICCRA)*, demonstrating their combined utility in analyzing adaptation effectiveness and resilience dynamics in real-world settings.



Causal chains for adaptation effectiveness and resilience

This Protocol presents a unified conceptual model to collect data on adaptation and resilience using causal chain thinking (Figure 1). The model builds on the logic and experiential grounding of impact pathways, which map interventions to outcomes and make explicit the assumptions linking them (Bosomworth et al. 2015; Haasnoot et al. 2019; Nowak 2023; Schuetz et al. 2017; Werners et al. 2021). It also draws conceptually from causal model research, emphasizing causal directionality and the role of different variable types in explaining causal linkages (Biesbroek, Dupuis, and Wellstead 2017; Diemer, Hudson, and Javaras 2021; Ferguson et al. 2020). A causal chain conceptual model, as used in this Protocol, is a structured representation that organizes plausible linkages among actions, conditions, and outcomes into a coherent sequence or pathway (FSIN 2014). It does not specify data requirements, or inference procedures; rather, it provides the conceptual foundations upon which analytical models are developed to test, estimate, and validate hypothesized connections.

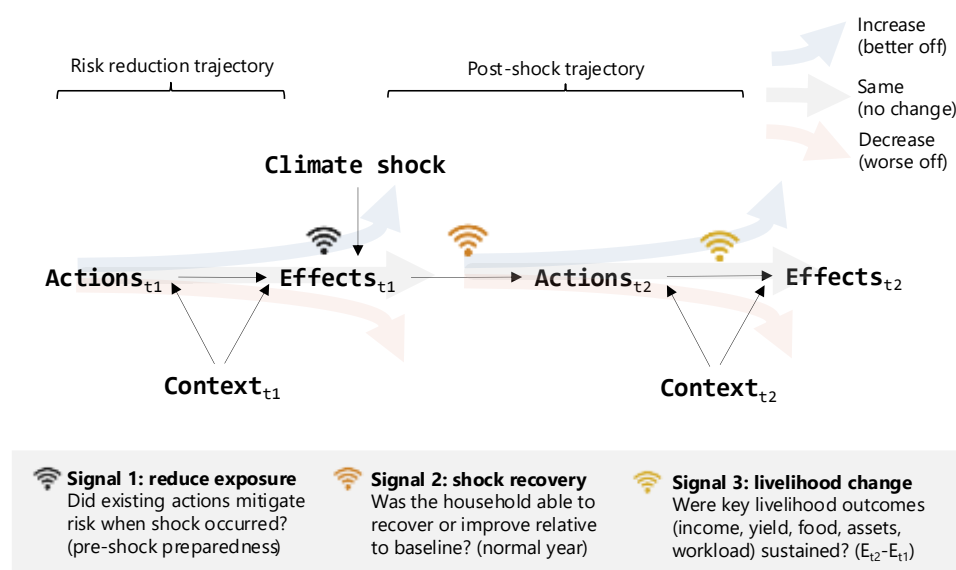


Figure 1. Adaptation effectiveness and resilience conceptualized as part of a dynamic adaptation process. Source: own.

The unified causal chain model conceptualizes adaptation as a dynamic process unfolding over time. At t_1 , households act before or during a shock or stressor (henceforth *shock*); these actions interact with context to shape immediate impacts (E_{t1}). At t_2 , new or continued actions—coping, adaptive, or transformative—again interact with context to determine recovery outcomes. The shock serves as an inflection point that resets baselines: effects observed at t_1 form the reference for assessing recovery at t_2 . The conceptual model distinguishes two trajectories—a risk-reduction phase before or during the shock, and a recovery phase afterward—showing that resilience cannot be captured through static outcomes but through changes relative to shifting baselines. Three resilience signals guide interpretation of adaptation outcomes. They capture



perceived or observable patterns that reveal how resilience manifests across time and outcomes. Risk exposure reduction compares losses at t_1 to pre-shock conditions; livelihood outcomes assess whether households sustain or improve yields, income, food security, or assets between t_1 and t_2 ; and recovery relative to baseline evaluates whether post-shock outcomes remain below, regain, or surpass pre-shock levels. Together, these signals indicate whether adaptation actions are reducing risk, supporting recovery, and strengthening preparedness for future shocks.

Causal chain components

Within this causal chain model, **actions** represent the diverse adaptation options and strategies that households adopt before and after shocks—preventative, adaptive, coping, or transformative¹. The combination of these actions, conditioned by household capacities and contextual thresholds, determines whether households' situation (e.g., livelihoods) stabilize, recover, improve, or conversely decline. This distinction between pre-shock and post-shock actions is a conceptual advancement of this protocol, as it helps operationalize the dynamic nature of adaptation. It recognizes that timing and sequencing of actions condition effectiveness and that strategies deployed before a shock may reduce (or not) losses, shaping risk reduction trajectories, while those adopted afterwards shape recovery trajectories.

Effects are the observed livelihood outcomes—such as yields, food security, assets, or income—that result from the interaction of climate shocks or stressors with adaptation actions and contexts. Because they are explicitly linked to both climate shocks and adaptation actions, these outcomes are interpreted as resilience outcomes rather than generic development gains. Livelihood outcomes can encompass a wide range of dimensions, including agronomic performance, economic stability, household well-being, or environmental conditions, and the specific indicators can be adjusted to match project goals or data availability. For example, yield stability under drought, household food access during lean seasons, or recovery of asset levels after flooding each provide different but complementary insights into how resilience manifests.

Context is treated here as a broad variable that, for simplicity, masks several important nuances. It encompasses the capacities, assets, and conditions that enable households to act and that also shape their recovery trajectories after shocks and stressors. Examples include financial resources, access to credit, social networks, information services, or functioning institutions. These elements open or constrain the space of possible responses, but they do not in themselves guarantee resilience outcomes. Moreover, context is not static: many capacities

¹ Preventative actions (e.g., soil and water conservation) are typically implemented before shocks, while adaptive adjustments (e.g., crop diversification) can occur before or after the shock. Coping actions such as distress sales or short-term labor migration, often occur in the aftermath of shock; transformative responses—like livelihood shifts or systemic changes—emerge when existing strategies are no longer viable (Béné et al. 2012; Eakin, Lemos, and Nelson 2014).



are influenced by external interventions—such as training, extension, or infrastructure—that expand options for farmers over time. In this sense, context both conditions the immediate effectiveness of actions and evolves through the very processes of adaptation that it enables.

The **arrows** in the figure depict two key effectiveness and resilience trajectories: (i) the risk reduction trajectory prior to or during a shock, and (ii) the post-shock trajectory that follows. These trajectories are essential because adaptation and resilience cannot be understood through static outcomes alone. A household's status at a single point in time tells us little about whether it avoided losses in the first place, whether it recovered, or whether recovery left it more or less prepared for the future. By following trajectories rather than snapshots, the causal chain model recognizes the dynamic nature of adaptation and resilience.

Resilience **signals** (icons) are measurable expressions of how adaptation unfolds over time. They translate the abstract idea of resilience into observable evidence of risk reduction, recovery, and livelihood change.

The first signal, effectiveness in reducing risk exposure, looks backward to assessing whether actions taken before or during a shock lessened its impact. In this sense, resilient households suffer smaller losses than they would have without such actions.

The second signal, recovery level, reflects how households have fared after a shock compared to a previous “normal”, whether they remain below, regain, or exceed pre-shock conditions. Resilience in this sense extends beyond returning to former or stable states, to an enhanced state despite disruption, which also ensures preparedness for future shocks.

The third signal, livelihood change, captures shifts in key outcomes such as productivity, income, or other well-being dimensions. Resilient households sustain or improve these livelihood outcomes under stress relative to E_{t-1} , while those experiencing losses become/remain vulnerable. This pathway captures resilience as continuity or improvement of livelihoods under stress, with E_{t-1} acting as the reset baseline imposed by the shock. These changes may be visible within larger timeframes—such as when incremental improvements in soil health or diversification only become apparent after several seasons.

These signals together offer flexibility to examine resilience across different timeframes. In contexts where shocks are recent or severe, the recovery signal may reveal resilience still in progress; in longer horizons, livelihood change becomes more visible—for instance, gradual gains from soil fertility restoration or livelihood diversification. The signals provide complementary insights into adaptation effectiveness, capturing how losses are reduced, recovery sustained, and livelihoods strengthened over time. Depending on data availability and program priorities, analyses may focus on one or more signals.



Scale of application

The Protocol is designed to capture data and **insights at the local level**—households and communities. The focus is on evaluating the effectiveness of interventions in strengthening resilience where climate impacts are directly experienced and managed. This local-level evidence is essential, as it provides the empirical basis for understanding how farmers respond to interventions and which pathways yield resilience outcomes. At the same time, the insights generated through this Protocol may feed upward, informing programmatic decisions at meso and national scales, and supporting programmatic or portfolio goals. In this way, the Protocol bridges the micro-level realities of farmer adaptation with the broader scaling and policy objectives of projects and programs, ensuring that agricultural adaptation investments are evidence-based and context-responsive.

Alignment with current approaches

This causal chain model builds on the growing consensus that resilience and adaptation effectiveness must be assessed in relation to specific shocks or adverse events—rather than in the abstract—because only under stress can adaptive capacity and recovery be meaningfully observed (Bahadur, Ibrahim, and Tanner 2013). It treats resilience as an outcome, in line with emerging conceptualizations that distinguish observable recovery or stability from the latent capacities that underpin them (Barrett and Conostas 2014; Clare et al. 2017; Pret et al. 2025). These capacities—absorptive, adaptive, and transformative—are recognized as fluid and overlapping, not fixed categories, and do not necessarily translate into resilience outcomes (Béné et al. 2012; Béné, Frankenberger, and Nelson 2015). Contextual moderators, long emphasized in the adoption and behavioral literature, explain why similar strategies yield different effects across settings. The model draws on extensive empirical and conceptual work showing that adaptation and resilience evolve across interlinked temporal scales—from immediate coping to gradual transformation—and emphasizes the importance of tracing these dynamic processes through time rather than capturing them as static outcomes (Béné 2013; Béné, Frankenberger, and Nelson 2015; Biesbroek, Dupuis, and Wellstead 2017; FSIN 2014).

The causal chain model in this Protocol advances these existing conceptualizations and approaches by providing a coherent causal structure that links adaptation actions, mediating processes, and resilience outcomes across time. It moves beyond static or compartmentalized assessments to show how effects unfold, interact, and accumulate under different conditions. By making these linkages explicit, it offers a shared logic for integrating diverse forms of evidence—empirical and experiential—and refining understanding as new insights emerge. Together, these contributions transform fragmented observations into a dynamic, cumulative picture of how adaptation generates resilience across contexts and timescales.



TOOLS FOR CAUSAL CHAIN DATA COLLECTION

This Protocol presents two complementary tools for collecting and analyzing data on adaptation effectiveness and resilience. The first is a **structured survey module** implemented at household level; it can be embedded in more detailed household surveys or implemented as a stand-alone tool. The **survey** systematically captures household-level data on resilience pathways across a large and diverse sample. While recall and perception data have limitations, they are often the only feasible way to document experiences of climate adverse events, strategies used, and outcomes such as recovery or food access. Structured survey questions ensure comparability across households and regions, making it possible to quantify patterns and test relationships at scale. The survey generates standardized variables that can be integrated into causal inference designs (e.g., matching, regression, mediation), enabling more robust analysis of what factors influence resilience trajectories. At the same time, survey-based recall questions anchor responses to concrete events (specific shocks, before/after actions), reducing ambiguity and helping separate pre-shock strategies from post-shock coping.

The second tool presented here is a **participatory focus group discussion (FGD) guide** conducted at the community level, designed to capture local causal chains, contextual mechanisms, and shared judgments of strategy performance. Few methodological tools exist for engaging farmers in causal analysis. Expert-based approaches, such as fuzzy cognitive maps (FCMs) or large system diagrams, have been used, yet these often become too complex, difficult to synthesize, or biased toward technical framings (Mehryar et al. 2019; Sarmiento et al. 2024). The FGD approach in this Protocol reduces such complexity by anchoring discussions in a temporal narrative: *then* (adverse climate event and livelihood impacts), *now* (current outcomes and practices), and *future* (anticipated outcomes and additional practices). This sequence mirrors the survey design and ensures comparability across sites while leaving space for contextual nuance. The use of causal inference logic, rather than exhaustive system mapping, allows diagrams to remain transparent, interpretable, and less prone to over-fitting. Importantly, participatory causal models elevate diverse sources of knowledge—especially those of women, elders, and other marginalized groups—thereby increasing the robustness and legitimacy of findings.

Each approach offers distinct advantages: surveys generate systematic, quantifiable evidence across large samples, while FGDs provide rich, place-based insights into mechanisms and probabilities that are difficult to observe otherwise. Depending on resources and objectives, users may employ one approach or combine them, with the latter offering the most complete and triangulated understanding of how adaptation strategies shape resilience outcomes.

Survey tool

This section guides enumerators and analysts in collecting data on adaptation effectiveness and resilience along the causal chain linking climate events,



household actions, and their effects. The module captures how households experience, respond to, and recover (or transform) from climate and weather shocks. It produces comparable evidence on risk exposure reduction, recovery relative to baseline, and livelihood effects under stress (Figure 2).

This data structure mirrors the conceptual causal chain model (Figure 1), where:

- A climate shock (S) occurs and produces immediate effects (E_{t1}), shaped by existing household context and pre-shock actions (A_{t1}).
- Households then take further post-shock actions (A_{t2}), which generate new effects (E_{t2}) during recovery.

The module can be used as a stand-alone instrument or integrated into larger household surveys. When used independently, it should be paired with basic household identification, demographic, and livelihood characteristics to contextualize results. We strongly recommend supplementing it with short questions on household assets and capacities (e.g., land size, savings, extension access, income diversification), which are critical for explaining differences in outcomes. Appendix A1 provides examples of how complementary modules can be leveraged for resilience analysis.

Core questions

Data collection is organized around the core building blocks of the causal chains, as outlined below. These questions are also illustrated in the annotated Figure 2.

Climate shocks (or adverse events)

Identify the **main** climate or weather event that affected the household in the past year, its **frequency** and **severity** compared to previous years.

Begin with a **short three-year scan** to establish exposure history, then focus on **one "most affecting" event** from the past year for detailed analysis.

Notes:

- ✓ A three-year scan captures exposure trends while minimizing recall bias from distant events.
- ✓ A one-year window offers better recall accuracy and temporal precision for linking actions and outcomes, improving reliability of self-reported data.
- ✓ If integrated into a larger survey, the same 12-month time anchor should apply across modules (e.g., yield or income data) to ensure internal consistency.
- ✓ The "most affecting" event is subjective; specify whose perspective it reflects and verify timing and type against the season to ensure consistency.

Effects of shock and recovery levels

Ask how the shock affected **key livelihood domains**—yields, income, food access, assets, and workload—relative to a normal year. This forms the impact baseline against which recovery from shock will be assessed.

Then, ask about **perceived recovery level** of the household.

Notes



- ✓ Livelihood domains might vary depending on the project/program, these will need to be adjusted in the survey.
- ✓ Consistent livelihood domains at E_{t1} and E_{t2} is important to ensure comparability across time and households.
- ✓ Perceptions of “normal” years may vary, introducing bias; enumerators should clarify what constitutes a normal year to respondents.

Pre-shock actions and effectiveness

Document preventive, adaptive or transformative ***practices already in place before the event*** (e.g., drought-tolerant varieties, mulching, savings, early planting, diversified income). These actions represent the household’s ex-ante adaptation portfolio and are important for identifying ex-ante adaptation capacity shaping subsequent outcomes (effects).

Next, ask about ***effectiveness of actions*** in reducing risk exposure - this becomes resilience signal #1.

Notes

- ✓ If this module is part of a larger survey, alignment with other modules (e.g., farm management) is important to maintain consistency and comparability
- ✓ Lists of actions should remain concise and align with the project/program intervention logic; grouped portfolios of actions can capture effectiveness efficiently, but might mask details (i.e., effectiveness will be interpreted in the context of the entire portfolio and not of individual actions)
- ✓ Respondents may over-report proactive behaviors; cross-check with observable or verifiable practices to enhance reliability.

Post-shock actions

Record coping and adaptive ***responses taken after the event*** (e.g., borrowing, asset sales, replanting, diversification, migration). Post-shock actions may be the same as pre-shock actions but might also include coping measures. These actions show how households mobilize resources when preventive measures are insufficient.

Notes

- ✓ Keeping pre- and post-shock actions distinct clarifies continuity, change, and learning processes, as it reveals whether households continue, scale up, or replace prior practices.
- ✓ When multiple shocks occur close together, respondents may conflate them or misattribute actions to the wrong event. Enumerators should probe carefully to establish timing and ensure that actions are linked to the focal shock only.

Effects post-shock (E_{t2})



Ask about ***changes in the same livelihood domains assessed at shock impact***—yields, income, food access, assets, and workload—to capture how households have fared since the shock. These indicators reveal whether the recovery trajectory has restored, improved, or further eroded household well-being.

Then, ask about ***perceived readiness for future shocks***. This hints to whether recovery has translated into sustained resilience or remains a temporary adjustment.

Notes

- ✓ Comparing identical livelihood domains (e.g., yield, income, food access, etc.) with E_{t-1} isolates both the degree and quality of recovery.
- ✓ Recovery perceptions can differ across household members; triangulating with quantitative indicators (e.g., yield or income data) strengthens inference

Context

Record information on household assets, land size, labor availability, access to information, credit, services, and social networks—factors that expand or constrain the space of possible responses. Context variables can help explain differential outcomes across households.

Note

- ✓ These variables are typically measured “as of survey time” and therefore might not be timestamped to specific shocks; but designers can choose to include them in the resilience survey module to capture dynamic in resource access or institutional conditions.

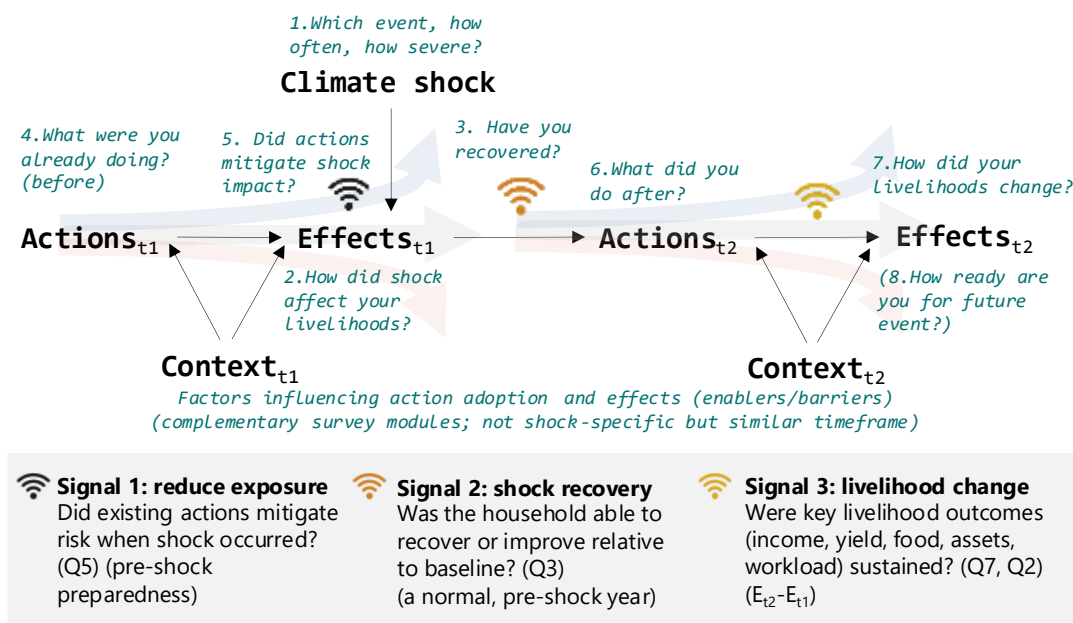


Figure 2. Causal chain elements (black text, colored icons) with core survey question annotations (blue italicized text). Source: own

Table 2. Reference guide to unpack causal chain elements and linkages

Question	Sub-questions
1. Which event, how often, how severe?	1.1. Which climate event have you experienced in the past 3 years? <i>(choose from list)</i> a. How often? (per event) <i>(1x, 2x, 3x,... never)</i> 1.2. Which event affected your household the most in the past year? <i>(rank from events listed in previous question)</i> a. When did you experience it? <i>(select months)</i> b. How was it compared to previous ones you faced in the past? <i>(less severe, same, more severe, worse ever)</i>
2. How did the shock affect your livelihoods?	2.1. How did the event affect the following a. Amount of crop harvested b. HH income c. Amount of work needed on farm d. HH access to enough and varied food e. Productive assets (livestock, seed stock, etc.) <i>(increased, did not change, decreased) (replace/add others)</i>
3. Have you recovered?	3.1. How is your HH doing now, compared to before the event? <i>(still struggling and not yet back to the level before the event; situation is even more difficult than it was before the event; we are back to living and farming about the same as before the event/ things did not change; we are now doing better than before the event)</i>
4. What were you already doing? (before)?	4.1. Before the event, which of the following strategies were you using? <i>(choose from list)</i>
5. Did actions mitigate shock impact?	5.1. Think about the strategies you were already using before the event,. How did they affect how badly your household was impacted by the event?



Question	Sub-questions
	<i>(made things worse-we were affected more because of the actions; made no difference-did not change how much we were affected; reduced how badly we were affected)</i>
6. What did you do after?	6.1. Which of the following strategies did you use after the event happened <i>(choose from list)</i>
7. How did your livelihoods change?	7.1. After the event and after using the strategies (remind which), how did the following change: a. Amount of crop harvested b. HH income c. Amount of work needed on farm d. HH access to enough and varied food e. Productive assets (livestock, seed stock, etc.) <i>(increased, did not change, decreased, too early to say)</i>
8. How ready are you for future?	8.1. Thinking about a similar or worse event next year, how prepared is your household to deal with it? <i>(would be badly affected again; could get back to how things are now; could recover and be better off then now; would make major changes to avoid future risks (off farm income, new crop, migrate)</i>

Sampling

Samples should be structured to enable comparison between households exposed and unexposed to specific interventions, while also capturing variation in shock exposure. Ideally, study sites (treatment and comparison areas) are identified *before data collection* based on clear eligibility or exposure criteria, followed by random selection of households from updated listings within each site. This design supports more credible causal inference, particularly when combined with statistical controls, stratification, or matching methods.

Where recent listings are unavailable, structured random-walk procedures can be applied, but these must be carefully parameterized to minimize bias. Starting points, household intervals, and replacement rules should be standardized and strictly supervised. Enumerators should avoid replacing unavailable households except under predefined conditions, as ad-hoc substitution introduces selection bias. Supervisors should conduct ex-post balance and representativeness checks to verify comparability across groups.

Design notes

Resilience needs to be assessed in relation to a clearly defined event following causal-chain logic. Each module anchors questions to a climate or livelihood shock, enabling the analysis of actions and outcomes relative to that event. Separating pre-shock and post-shock actions operationalizes adaptation as a dynamic process rather than a fixed condition, clarifying how households prepare for, experience, and recover from shocks.

Question order is critical for reducing bias and preserving causal clarity. Shock effects and recovery levels are elicited before pre-shock actions to avoid priming



and justification bias. This sequencing ensures that outcome reporting remains independent and allows triangulation between actions and effects during analysis.

Most questions generate qualitative information that can be standardized for analysis. Responses are captured through relative scales or categorical options rather than exact figures, reducing recall burden. These data can be numerically coded to support quantitative analysis while retaining contextual nuance. The approach balances precision with feasibility: it allows consistent measurement across respondents without expecting recall of specific values. Its main strength is realism in low-data settings; its limitation lies in cross-context comparability, which depends on consistent coding and clear reference points.

Adoption is treated as a mediator, not an outcome. External interventions—such as climate advisories, training, or CSA hubs—are conceptualized as enabling conditions that increase the likelihood of adoption. Adoption mediates between exposure and livelihood effects, clarifying causal direction and distinguishing enabling influences from direct outcomes.

Implementation notes

Rigorous preparation and adherence to question sequencing are essential for data quality. Enumerators must be trained to elicit outcomes before actions, and to clearly distinguish between effects at impact (E_{t+1}) and recovery outcomes (E_{t+2}). A minimum of one week of training is recommended, combining classroom sessions, role plays, and pilot testing.

Training should emphasize local interpretation of shocks. Terms such as “drought” or “dry spell” vary across communities; enumerators must learn local equivalents and test comprehension. Piloting should assess not only understanding but also recall accuracy, fatigue, and skip pattern functionality.

Digital data collection ensures consistency and real-time quality control. Platforms such as KoboToolbox or ODK help standardize response options and enable daily checks. Supervisors should conduct spot checks and review submissions each day to detect inconsistencies, high “don’t know” rates, or evidence of fatigue. A feedback loop should be maintained to correct enumerators immediately.

Module placement within longer surveys affects data quality. When embedded in a multi-topic questionnaire, the resilience module should be positioned after basic household questions but before technical or cognitively demanding sections. Field experience (AICCRA Ghana) showed that mid-survey placement sustained respondent engagement. Interview duration should be monitored closely; when integrated into larger household surveys, analysis should be limited to a single focal shock to keep total interview time under two hours.

Triangulation during and after strengthens interpretation of subjective resilience measures. Households may report strong buffering but weak recovery, or recovery alongside low confidence for future shocks. Capturing these nuances provides a layered understanding of resilience as an evolving process rather than



a single metric. Field teams should cross-check household responses across related questions and with observable indicators whenever possible. During analysis, triangulating perceived outcomes with quantitative measures—such as yield, income, or asset data—helps validate reported resilience patterns. This approach ensures that resilience is captured as a dynamic process, not a single perception or score.

Interpreting resilience signals

Resilience cannot always be captured through fully realized outcomes, as many effects of adaptation unfold gradually or remain latent until future shocks occur. Focusing on resilience signals—early, observable indications of change—allows analysts to detect whether systems are moving toward greater stability, adaptability, or transformation even before final outcomes manifest. Table 3 provides a reference guide for identifying, interpreting, and triangulating such resilience and effectiveness signals.

Table 3. Quick reference guide for interpreting resilience signals

Indicator	Analysis & Interpretation	Cross-checks
Risk exposure reduction (Q5)		
% of HH reporting reduced risk exposure, given actions taking prior to shock <i>(by HH type, shock type, shock intensity)</i>	Reflects buffering capacity of actions already in place at the time of the shock. Resilient HH experienced “reduced risk” after adopting actions, meaning also effective adaptation; “no effect” and “increased risk” signal sustained/ increased vulnerability and ineffective/ maladaptive actions.	Q2 (shock impacts/ E_{t1}) to test partial buffering (e.g., experienced decreased yields but actions helped to reduce risk = actions were partially effective) Q3 (recovery) to see if risk exposure translated into recovery (e.g., HH reporting reduced risk exposure may also be more likely to judge themselves “same as before” or “better off.”
Recovery (Q3)		
% of HH reporting recovery after shock <i>(by HH type, shock type, shock intensity)</i>	Summarizes the HH’s general sense of being “back on their feet” after an intense shock. Resilient HH report recovery to baseline “same as before” or improved conditions (“better off”); “partially” (below baseline) and “not recovered” signal sustained/ increased vulnerability. Can also be interpreted as a continuum (left-right: not recovered-partial-recovered-better off)	Q2 (shock impacts/ E_{t1}) to test if more damaging events reduce chances of recovery). Q7 (post-shock outcomes/ E_{t2}) to examine if feeling of recovery translates into livelihood enhancements. Q8 future preparedness to examine linkage between experienced and potential resilience
Livelihood changes after shock (Q7 vs Q2)		



Indicator	Analysis & Interpretation	Cross-checks
% of HH reporting resilient livelihoods (by HH type, shock type, shock intensity, livelihood domain)	Summarizes how wellbeing domains evolved during recovery (trajectories from shock impact (Et1) to post-shock outcomes (Et2), whether HH conditions stabilize, improve or erode since shock impact. Resilient HH move from losses or stable outcomes to gains (they climb back and/or improve after their lowest point), i.e., decrease→ increase; no change→no change/increase; Trajectories from increase → decrease or decrease→ no change suggested eroded livelihoods (further decline). Possibility to compute resilient livelihood score across domains of interest.	Q6 (post-shock actions) to identify which responses explain stabilization or improvement. Q3 (generic recovery) to test alignment between perceived and observed outcomes. Other modules (capacities) to test whether capacities translate into observed outcomes
Aggregate resilience signal (optional)		
Composite index (0–3) of HH resilience signals	Summarizes multidimensional resilience in a single score. Useful for portfolio-level reporting or comparison across contexts. Assign a positive score for “resilient” responses across each signal. Sum can be weighted depending on program priorities; e.g., programs emphasizing anticipatory adaptation may assign higher weight to risk exposure reduction, while those focusing on long-term wellbeing may weight recovery outcomes more heavily. Results to be interpreted in relation to other modules/signals, else they mask nuances.	Use sensitivity tests to check robustness of thresholds/weights. Report distributions across lenses as well as the index to preserve nuance.

Caveats

Data from the survey module are based on household perceptions, which introduces subjectivity and social desirability bias; subjectivity in such assessments is both inherent and desirable, as the goal is to understand lived experiences (Béné et al. 2019; Clare et al. 2017; Jones and Tanner 2017). Respondents may overstate recovery to present themselves in a positive light or understate it to highlight vulnerability. These biases can be mitigated by sequencing (eliciting outcomes before actions), triangulating with other survey modules, and validating against external datasets (climate, ag census, etc.).

Recall bias is also an inherent problem when asking for information about the past. Respondents may struggle to remember the details of past shocks or may conflate different events. The design addresses this by narrowing to a single focal event in the past 12 months and by using local calendars, seasons, or festivals to anchor recall. However, some errors remain unavoidable, especially for prolonged events such as droughts or dry spells.

Survey length and fatigue can compromise data quality. When resilience modules are embedded in large multi-topic surveys, interviews often exceed two hours. Fatigue may reduce the accuracy of responses or increase refusal rates. In AICCRA, these risks were mitigated by trimming modules, limiting resilience



analysis to a single focal event, and repositioning the module mid-survey. Nonetheless, implementers should be aware that trade-offs exist between depth of data and respondent burden.

Finally, the module captures outcomes over a relatively short recall period and may miss longer-term transformations. It is best suited to measuring short- and medium-term adaptation effectiveness and resilience trajectories rather than systemic change. For this reason, combining it with qualitative approaches such as focus group discussions or longitudinal studies is recommended.



Focus group discussions (FGDs)

FGDs are designed to complement household survey data by unpacking the causal chains that link climate shocks, livelihood impacts, practices, and resilience outcomes; or to substitute survey where resources for large-scale data collection are limited and when the priority is collecting qualitative insights. Using the logic of causal chains (Figure 1), FGDs explore how farmers perceive the “what” (actions), “why” (underlying factors), and “so what” (effects) of their adaptation actions. The discussions generate evidence that can be summarized qualitatively for context or translated into quantitative structures—such as causal graphs or Bayesian networks—that capture directionality, relative importance, and heterogeneity of pathways. While causal diagrams are traditionally elicited from technical experts, here farmers themselves are treated as experts of their lived experience. The FGD guide in Table 3 operationalizes this innovation by enabling farmers to articulate their knowledge of resilience pathways in ways that are structured yet accessible.

Core questions

The following guiding questions (Table 3) provide a structured yet flexible pathway for discussion. In Box 2, we provide recommendations on potential adjustments required when climate events are recent. The guiding questions are organized to follow the temporal flow of shocks and responses—before, during, and after a climate event—while leaving room for participants to share diverse experiences. Together, these sections help reconstruct resilience pathways and the mechanisms that shape them:

- **Livelihoods:** captures baseline context and links livelihood sources to resilience
- **Climate events (risk exposure):** identifies common shocks and distinguishes frequency and severity
- **Impacts (shock effects):** explores immediate effects of the most significant event, including heterogeneity across groups
- **Recovery and livelihoods:** assesses post-shock situations and the diversity of recovery trajectories
- **Actions and effectiveness:** documents strategies used, clarifies timing (before/after), and probes effectiveness
- **Other mechanisms:** surfaces enabling or constraining factors beyond household practices
- **Future resilience:** probes anticipation, lessons learned, and preparedness for future events

Table 3. Guiding questions and instructions for FGDs

Guiding questions (orientate)	Facilitator instructions
A. LIVELIHOODS	
1. What are the main livelihood sources in this community? Which crops and animals are most important?	<i>Refer to farming and non-farming activities; understand reliance in times of shock</i>



Guiding questions (orientate)	Facilitator instructions
2. For farming: is the main purpose consumption, selling, or both? Which products are sold most often, and why?	Probe main income vs. food sources; gender roles in production/marketing
B. CLIMATE EVENTS (EXPOSURE)	
3. What climate or weather problems do you usually face in the community?	Encourage reflection on changes in rainfall, temperature, or seasons over the years
4. Of all these, which one was the most serious in the last 3 years? Why?	Probe severity vs. frequency; discuss impacts felt across households
5. When did this event happen (which year/season)? Who was most affected, and why? What did you experience?	Probe timing of event, groups affected, and context-specific details (duration of dry spell/ drought; when rain stopped, how hot it was; how high was the water)
C. IMPACTS (IMMEDIATE EFFECTS OF THE SHOCK)	
6. What impacts did this event have on your farming and livelihood?	Probe: crops, income, food, work, assets
7. Did everyone in the community experience the same impacts? If not, why not?	Probe: wealth, land, gender, timing of the event, location
8. Right after the event, what did you do to cope?	Probe: reduced food, borrowed money, sold assets, migrated, etc.
D. RECOVERY & LIVELIHOOD EFFECTS	
9. Compared to when the {climate event} happened, how is your situation now - better, worse or the same? How many of you think this way? Why do you think some farmers see different results than others?	Probe: wealth, land, gender, access to Inputs, timing, severity of event
10. What specifically got better compared to last {climate event} and why? Who experienced this improvement?	Probe for drivers of improvement (practices, support, conditions)
11. What specifically got worse, and why? Who experienced this decline?	Probe reasons: crop failure, lack of recovery, continued stress
12. What has stayed the same, and why? Who experienced this stability?	Probe persistence of conditions, resilience factors
E. ACTIONS & EFFECTIVENESS	
13. Thinking of the event and the outcomes we just discussed, what actions or practices did you use to address the impacts?	Allow free listing—farm practices, income strategies, information use, etc.; try to capture how many practices each
14. For each action mentioned: Was this something you were already doing <i>before</i> the event, or that you started <i>after</i> ?	Clarify timing and intensity; note continuity vs. change
15. Which of these actions do you think helped you most to achieve a better outcome? Why?	Probe to understand if and how actions already in place before the event reduced/magnified how badly they were affected when event hit
16. Which actions were less effective, or even harmful? Why do you think so?	
F. OTHER MECHANISMS	
17. Apart from your own practices, what other factors helped you deal with the event?	Probe: credit, inputs, extension, community support, gov/NGO help.
18. What prevented you from achieving better results?	Probe: lack of resources, information, labor
G. LEARNING & FUTURE RESILIENCE	
19. From this experience, what did you learn that has changed how you farm or prepare for the future?	Probe for lessons, adjustments, shifts in behavior
20. If a similar event happened again soon, do you think you would be better off, worse off, or about the same? Why? What else would need to change for you to be better prepared?	Probe for confidence, enabling conditions, support needed

Box 2. Guidance for mapping recent or ongoing climate events

Acknowledge the timing: If participants describe the current season as the worst climate event, clarify that you are interested in both their current experiences and expectations for recovery.



Differentiate impacts vs. outcomes: Ask about what has already happened (e.g., crop damage, reduced food, income loss) separately from what is still unknown (e.g., final harvest, market sales). Record clearly when an outcome cannot yet be judged.

Capture anticipation: If recovery is ongoing (e.g., crops still in the field), ask participants: “How do you expect things will turn out by harvest time?” Encourage them to explain what their expectations depend on (e.g., rainfall continuing, pest control, market conditions).

Mark uncertainty explicitly: Use categories like “too early to say” or “uncertain” when farmers cannot provide recovery outcomes. This aligns with survey coding (e.g., J3.2 “too early to know”).

Probe for coping vs. recovery: Even if final recovery is unknown, farmers can still describe immediate coping strategies (skipped meals, borrowing, selling assets) and short-term changes. Ensure these are captured as distinct from longer-term recovery.

Allow multiple reference points: If participants prefer to talk about an earlier, completed event instead of the current one, let them do so, but note clearly which event the discussion refers to.

Maintain flexibility: If time allows, you can ask participants to discuss both the ongoing event (current impacts and expectations) and the last completed event (observed outcomes). This gives a richer picture without forcing them to speculate beyond their comfort.

Sampling

Communities are selected purposively using maximum-variation criteria: agro-ecological diversity, exposure to different climate risks, and levels of access to information or interventions. Within each community, two FGDs are conducted (typically one with women, one with men), each involving 7–9 participants. The aim is thematic saturation rather than statistical representativeness. Stratification by gender is critical to ensure that distinct experiences, constraints, and mechanisms are captured. Participation is voluntary and ethical considerations (confidentiality, informed consent) are respected throughout.

We suggest elicitation methods that do not require the participants to have scientific or technical literacy (including mathematical graph theory underpinning DAGs); or literacy at all; we use dialogue/discussion and visual cues; these “maximize the legitimacy of the participant, minimize the opportunities for response biases, and do not rely on deep scientific or technical literacy of the participants” (Davies 2023)

FGD Implementation

FGDs conducted **after the survey** allow researchers to ground abstract patterns in lived experiences. Survey data often reveal correlations or broad trends but lack detail on the mechanisms driving them. Post-survey discussions provide the depth needed to interpret these findings, uncover why certain practices worked for some households but not for others, and identify contextual moderators such as gender, wealth, or timing of shocks. They also help validate directionality in causal chains and highlight unanticipated mechanisms or trade-offs not captured in the survey.



The main limitation is that survey instruments must already be finalized, which means that if relevant practices or outcomes were omitted, FGDs can only document them qualitatively without integrating them into the comparative dataset. In addition, there is a risk of confirmation bias, as facilitators may steer discussions too closely to survey results rather than leaving space for new insights.

By contrast, FGDs conducted **before the survey** can shape the design of the questionnaire itself. Early discussions reveal locally relevant shocks, outcomes, and practices, ensuring that the survey reflects community priorities and uses context-appropriate language. This is particularly valuable in new study areas or when working with marginalized groups whose perspectives are often overlooked in standardized survey tools.

The drawback of this approach is comparability; surveys tailored to each site may limit cross-site analysis, reducing the ability to aggregate results or benchmark across interventions. Pre-survey FGDs also risk overemphasizing practices that are highly salient in group discussions but less representative at population level. Finally, because the survey is not yet in place, findings from early FGDs may remain anecdotal if not systematically followed up through structured measurement.

Duration: Sessions should last no more than 1.5 hours to minimize fatigue, though exploring multiple shocks may require more time. The guiding questions are deliberately concise to respect participants' time, but expandable when conditions permit.

Facilitation: A team of at least three is recommended: facilitator, note-taker, and translator. This arrangement ensures both quality and inclusivity of data collection. The facilitator guides the discussion and manages group dynamics, while the note-taker documents non-verbal cues, disagreements, and contextual details that may be missed in transcripts. The translator bridges language barriers and helps adjust technical phrasing to culturally appropriate terms. Having these distinct roles also reduces cognitive overload on the facilitator and increases accuracy. Audio-recording, where consent is granted, provides an additional layer of reliability by allowing cross-checking of notes and minimizing the risk of selective capture.

Separate FGDs for women and men help reveal differentiated mechanisms, but cross-group synthesis is critical during analysis.

Where possible, facilitation should rely on visual prompts (e.g., pre-printed cards with illustrations of climate events, practices, and outcomes) placed centrally so that all participants can see and react. Cards provide both structure and inspiration for brainstorming. Unlike flipcharts—which often become illegible or exclude participants who cannot read—visual cards maximize inclusivity and reduce response bias. Cards or tokens should be adapted to local contexts. Space is needed to lay them out visibly on the ground or central table.



Discussions typically begin with mapping outcomes to avoid privileging practices upfront, but facilitators must remain flexible, allowing the sequence to adapt to participants' logic.

Analysis and interpretation

Using FGD data independently

When resilience pathway data are collected only through FGDs, the results should be interpreted primarily as *perceptions of causal relationships* rather than measured outcomes. Farmers' narratives offer valuable insight into how shocks are experienced, which practices are considered effective, and what enabling or constraining factors shape recovery. The emphasis should be on mapping *resilience* pathways (e.g., coping → outcomes → future preparedness) and identifying variation across groups (men vs. women, wealthier vs. poorer households).

Analysts should avoid over-quantifying responses. The strength of stand-alone FGDs lies in the richness of contextual explanations and the diversity of pathways described.

Transcripts or detailed notes are first coded against the guiding causal chain model (Figure 1): climate events or shocks, effects (E_{t1} , E_{t2}), actions (A_{t1} , A_{t2}), other mechanisms, and future resilience. Within each code, responses can be summarized into themes (e.g., "soil fertility practices reduced impacts of drought" or "women had less access to credit for recovery"). Analysts can then create causal pathway matrices, where rows represent climate events and columns represent outcomes, with linked practices or mechanisms recorded in each cell. Divergences across groups (men vs. women, wealthier vs. poorer) should be coded explicitly. Counts (e.g., number of participants raising hands for a practice) can be tabulated, but only as indicators of salience or consensus, not statistical representativity. The end product is a qualitative causal model, potentially simplified into a diagram (nodes = shocks, actions, effects; arrows = farmer-perceived linkages).

The main caution is that such evidence reflects subjective experience, which may be shaped by recall bias, social desirability, or dominant voices within groups. Transparency in documenting these dynamics, and triangulating across communities, increases robustness.

Using FGD data to complement the survey

When FGDs are paired with survey data, their role is to *deepen interpretation*. The resilience survey module quantifies exposure, impacts, strategies, and recovery outcomes, but cannot by themselves explain why some practices worked or under what conditions. FGDs fill this gap by unpacking mechanisms, validating directionality of causal chains, and surfacing missing variables (e.g., informal credit, community norms). For example, if survey results show that adoption of a certain practice is associated with higher recovery, FGDs can clarify whether



farmers attribute this to the practice itself, to access to resources, or to broader contextual factors.

Analysts should resist forcing FGDs into confirmatory evidence; their value lies in illuminating *how* and *why* survey patterns occur, and in highlighting heterogeneity or contradictions.

When paired with survey data, FGD coding follows the same structure but is aligned to survey variables. For example, if the survey includes a question on recovery of food security, FGD notes on “improved food access” should be coded under the same category. This enables side-by-side comparison: the survey shows the proportion of households reporting recovery, while FGDs reveal why recovery was possible or constrained. Practices mentioned in FGDs should be coded as “pre-shock” or “post-shock,” matching survey question, and flagged for whether participants considered them effective. Where FGDs identify new factors not in the survey (e.g., community labor sharing), these should be logged in a separate “emerging factors” category to inform interpretation and future survey design.

A key caution is comparability: if new practices or outcomes emerge in FGDs that were not captured in the survey, these can enrich interpretation but cannot be retroactively quantified. Instead, they should be treated as hypotheses or recommendations for future data collection.

FGD Limitations

Eliciting causal chains from community discussions is not without challenges. *Establishing baselines (“normal” versus “post-shock” conditions) takes time and may be contested.* Relationships are rarely linear; multiple, interacting shocks complicate attribution. Effects vary significantly across social groups, meaning that some mechanisms may be masked if only majority views are captured. Finally, while participatory causal models provide invaluable insights, they remain perceptions rather than directly measured effects; their strength lies in complementing, not substituting, household survey or observational data.

Farmers may struggle to distinguish between “normal” conditions and post-shock states, particularly when shocks recur frequently. This can blur the starting point of a causal chain. Facilitators should anchor questions in specific reference years, seasons, or major events and use visual prompts to differentiate time periods, thereby creating a clearer baseline for comparison.

Outcomes are rarely linear—small changes in practices may lead to large effects, and vice versa. This complexity can frustrate efforts to construct clear causal chains. To manage this, facilitators should probe for thresholds (e.g., “when did the practice start making a difference?”) and highlight instances of interaction between shocks, practices, and outcomes, ensuring that the final model captures non-linearity without becoming unmanageably complex.



Effects often vary across wealth groups, gender, or land access, meaning majority views can mask important differences. Stratifying FGDs by gender (and, where feasible, other characteristics such as age or livelihood orientation) allows marginalized voices to surface. Facilitators should explicitly ask about divergent experiences and encourage participants to discuss why outcomes differ across sub-groups.

FGDs rely on perceptions, which are shaped by recall bias, social desirability, group dynamics, and facilitator influence. Recall bias can be mitigated by anchoring questions in concrete events (e.g., a specific drought year). Social desirability bias can be reduced by ensuring confidentiality and framing questions neutrally (e.g., avoiding “good” vs. “bad” practices). Power dynamics within groups should be managed by encouraging quieter participants to speak, while facilitator bias can be minimized through training, standardized guides, and triangulation with survey and observational data.



USE CASE: AICCRA

Accelerating Impacts of CGIAR Climate Research for Africa (AICCRA) is a World Bank-funded Project supporting the scale-up of climate-smart agriculture. AICCRA works through partnerships in West Africa (Senegal, Mali, Ghana), East Africa (Ethiopia, Kenya), and Southern Africa (Zambia) to scale climate-smart innovations that help farmers anticipate, absorb, and adapt to climate risks. AICCRA has been implemented in two phases, 2022-2023 and 2024-2025.

The countries included in the AICCRA project are among the most climate-vulnerable regions in Africa, facing recurrent **climate risks** such as droughts, dry spell, erratic rainfall, flooding, and rising pest and disease pressures that undermine agricultural productivity and household food security. These risks translate into yield variability, income instability, and increased vulnerability to poverty, particularly among smallholder farmers whose livelihoods depend on rainfed agriculture.

Activities promoted by AICCRA have been tailored to local priorities but share a common focus on strengthening climate information services, agri-advisory systems, integrated climate-smart agriculture (CSA) practices, and sees system strengthening. In many cases, these are promoted as bundles to maximize synergies and adoption; for example, climate information paired with advisory services, or pest management combined with improved seed and soil fertility practices. For example, in Ghana and Senegal the project emphasizes integrated pest and soil fertility management alongside climate and weather information services; in Ethiopia and Kenya it strengthens climate-smart seed systems and digital advisories, while in Zambia and Mali it supports water management, agroforestry, and farmer-led extension models. These interventions are implemented with national agricultural research systems, extension services, private sector actors, and farmer organizations to ensure local ownership and long-term sustainability.

In Ghana, where this Protocol has been tested, AICCRA has been scaling climate-smart integrated pest management (CS-IPM) and complementary CSA practices through a combination of radio advisories, extension services, and Climate-Smart Agriculture (CSA) hubs. Radio is used as a low-cost, wide-reaching channel to deliver localized weather forecasts, seasonal advisories, and pest/disease management messages in local languages, while extension services and CSA hubs provide hands-on demonstrations, farmer training, and peer-to-peer learning. These approaches are deployed across six regions—Upper West, Upper East, Northern, Bono East/Brong Ahafo, Central, and Greater Accra—which together represent a diversity of agroecological zones, from drought-prone savannahs in the north to more humid, market-oriented systems in the south. By strengthening farmer access to timely, actionable information and linking mass outreach (radio) with personalized support (extension and hubs), AICCRA Ghana aims to enable smallholder farmers—especially those cultivating maize, cowpea, tomato, yam, and sweet potato—to better anticipate and manage climate risks, recover more



effectively from shocks, and increase productivity, while also addressing gender and inclusion gaps in access to services and technologies.

The theory of change underpinning AICCRA assumes that if farmers and value chain actors gain access to actionable climate information, improved technologies, and enabling institutional arrangements, then they are more likely to adopt climate-smart strategies that stabilize yields, protect natural resources, and reduce vulnerability to shocks. Over time, these practices are expected to contribute to enhanced resilience, food security, and inclusive growth. The project also emphasizes gender and social inclusion, recognizing that women and marginalized groups face specific barriers to accessing services and technologies. By building capacities at individual, institutional, and system levels, AICCRA aims to generate outcomes that extend beyond project sites, informing national climate strategies and contributing to the global evidence base on adaptation effectiveness.

Survey implementation

In the AICCRA project, the **resilience module** was embedded within a broad, multi-topic household survey designed to capture the multiple dimensions of agronomic practice, livelihood, and well-being (See Appendix 1). The survey encompassed modules on household identification and roster, land tenure, asset ownership, plot use and labor arrangements, input application, integrated pest management (IPM) and associated health effects, climate information services (CIS) and climate-smart agriculture (CSA) practices, dietary diversity, food security, off-farm income, and access to services and social protection.

While the wider survey provided **breadth of coverage**, the resilience module introduced **temporal depth**, explicitly linking pre-shock and post-shock states of households. This temporal framing enabled the testing of *adaptation effectiveness* against specific climatic or biophysical shocks—an analytical feature rarely attainable from standard cross-sectional modules.

The design of the resilience module ensured **internal consistency** across survey components. Lists of pre- and post-shock actions were intentionally harmonized with practices captured elsewhere—particularly in Module H (Integrated Pest Management) and Module I (Climate Information and CSA Practices). This alignment enables **cross-module triangulation**: analysts can validate whether practices households reported using in the resilience section are consistent with those reported as adopted or learned in other parts of the survey. It also supports testing whether the *perceived effectiveness* of actions can be plausibly linked to AICCRA interventions, such as exposure to radio advisories or CSA hubs (Appendix 1).

Beyond practice-level validation, triangulation extends to **capacity dimensions**. For instance, a household reporting the use of seasonal forecasts or extension advice in Module I can be compared with its recorded pre-shock preparedness in the resilience module. Likewise, pest management practices listed under Module



H can be examined against reported outcomes on harvest, pest losses, or income. Such cross-checks are vital for disentangling resilience outcomes that arise from *household agency* versus those enabled by *external interventions*. They strengthen causal interpretation by ensuring that resilience is measured not as a single attribute, but as the *emergent product of interactions* between shocks, capacities, actions, and effects.

In Ghana, **sampling combined pragmatic field realities with quasi-experimental rigor**. Control communities were identified by agricultural officers following strict criteria—similar socioeconomic characteristics and a minimum 10-km buffer from AICCRA-supported sites—to minimize contamination. Within communities, incomplete or outdated household listings made full sampling frames prohibitively expensive to construct; hence, structured *random-walk procedures* were applied under closely supervised protocols. Although less precise than complete listings, this approach ensured broad coverage across agro-ecological zones and intervention types. Ex-post balance tests confirmed acceptable comparability between treatment and control groups, supporting credible inference on resilience outcomes.

FGD implementation

Focus group discussions were conducted in twelve communities: eight with AICCRA interventions (four engaged early in the project and four in its second phase) and four without project exposure. This configuration enabled comparison across varying levels of program engagement. The sites represented distinct agro-ecological zones—notably the drier Upper West Region and the more humid and market-connected Bono Region—allowing exploration of how local environmental and institutional contexts shape adaptive responses.

Separate FGDs were organized for **women and men** in each community to ensure inclusive participation. A mixed-gender pilot revealed limited women's engagement, prompting gender-segregated sessions that substantially improved participation quality. Each community hosted two FGDs (7–10 participants per group), summing to approximately 230 participants overall.

In AICCRA's participatory design, **visual card-based elicitation** proved transformative. The cards, entirely pictorial and context-specific, depicted shocks, actions, and outcomes relevant to local farming systems. They served multiple purposes: structuring discussion, stimulating recall, and bridging literacy gaps. Women participants, in particular, responded with greater confidence and engagement when visual cues were introduced. The process, however, was time-intensive—typically lasting about two hours—and required sufficient space to display cards on the ground in a central circle visible to all participants. Despite these logistical demands, the approach yielded richer, more nuanced insights into farmers' reasoning and adaptive strategies.

FGDs were conducted **after the survey phase**, providing a platform to interpret and deepen quantitative findings. They helped unpack distinctions that structured



survey data alone could not capture—for example, the difference between *dry spells* and *droughts* in farmers' perceptions, or why households reporting higher yields did not necessarily experience higher incomes, often due to market access or price constraints. Discussions also clarified mechanisms behind *slow or incomplete recovery*—such as limited access to quality seed, unreliable weather forecasts, or low trust in information sources. These contextual insights revealed the complex interplay between knowledge, access, and enabling conditions that determine whether adaptive practices translate into real resilience gains.

Overall, the FGDs complemented the survey by transforming static data points into **dynamic causal narratives**, illuminating how households perceive, act upon, and learn from climatic disturbances within broader systems of constraint and opportunity.



CONCLUSION

This working paper introduced a conceptual model and operational protocol for collecting causal-chain data to assess adaptation effectiveness and resilience in agricultural systems. Together, the model and its two tools—a structured household survey module and a participatory focus group discussion (FGD) guide—demonstrate how complex adaptation processes can be systematically observed, quantified, and interpreted. Each tool brings distinct strengths: the survey module offers structured, comparable data across scales and shocks, while the FGD approach provides contextual depth, uncovering mechanisms and conditional factors that quantitative instruments often overlook. Their combined use allows triangulation between self-reported experiences, observed actions, and inferred outcomes, resulting in a richer, more coherent picture of how adaptation unfolds in practice.

At its core, the protocol advances the understanding that neither adaptation effectiveness nor resilience can be inferred from the presence of practices or capacities alone. Capacities such as credit access, extension services, or asset ownership are enabling conditions, not guarantees of resilience. Likewise, the effectiveness of adaptive actions depends on their timing, intensity, and interaction with specific shocks and thresholds. This causal chain model therefore repositions resilience measurement around causal logic—anchored in climate rationale, informed by context, and grounded in outcomes that reflect lived recovery and stability rather than assumed potential. By linking shocks, actions, and effects through a consistent temporal and relational lens, the protocol transforms fragmented snapshots into evidence of dynamic system behavior.

A key contribution of this effort lies in bridging monitoring, evaluation, and impact assessment (M&E and IA). Traditionally, these domains have operated in parallel—monitoring focusing on activity tracking, evaluation on outcomes, and impact assessment on attribution. The causal-chain model brings them together within a single empirical logic, enabling systematic data collection that serves all three purposes. When the household survey module is implemented regularly, it can track resilience trajectories over time, reducing reliance on recall and capturing adaptation as an evolving process rather than a static state. For impact assessments, the model strengthens causal inference by linking exposure, capacity, and outcomes along explicitly defined pathways. In doing so, it provides a shared backbone through which M&E systems and research evaluations can generate cumulative, comparable, and decision-relevant evidence.



Looking ahead, the next frontier lies in scaling this conceptual causal model through smarter and lighter data systems. Future applications should explore how core context variables, such as land use, vegetation stress, or exposure gradients, can be captured using remote sensing and climate and spatial data integration, reducing respondent burden and improving temporal coverage. Equally, modelling approaches that combine multiple data sources can further enrich causal understanding while minimizing respondent burden. By addressing the long-standing fragmentation of adaptation evidence, this protocol marks a step toward a more unified science of adaptation effectiveness, that is empirically rigorous, operationally feasible, and designed to learn continuously from the dynamic realities of climate resilience.



APPENDIX

Appendix 1. AICCRA Survey: Assessing linkages between capacities, context, and resilience outcomes

The table shows how each survey module can contribute to analyzing resilience by linking household capacities, strategies, and conditions to the outcomes captured in the dedicated resilience section (J). Some modules provide **capacities** (e.g., assets, land, services), others capture **direct outcomes** (e.g., food access, dietary diversity), and some provide **contextual information** (e.g., household characteristics). Together, they open multiple analytical pathways: from simple associations that reveal patterns, to more advanced causal designs that test pathways of influence. Other modules also support **data triangulation**. For example, modules H (Integrated pest management) and I (Climate information and CSA sources) provide detailed data on which practices and information households had access to and used. These can be triangulated directly with the resilience module (pre-shock strategies, post-shock actions) to validate whether the strategies farmers report in the resilience module match their adoption and information sources recorded elsewhere in the survey. This cross-check strengthens confidence in reported behaviors and helps distinguish knowledge from actual uptake. The examples in Table A1.1 illustrate just a few ways these modules can be tapped; in practice, there are many possible analytical approaches possible depending on the research question.

A key aspect in interpreting the survey is timing relative to climate shocks. Most sections (A–I, K–O) capture conditions at the time of the survey or during the most recent agricultural season. These are valuable for understanding the resources and strategies households have, but they cannot confirm whether those capacities were already in place before a specific shock or developed afterwards during recovery. For example, an asset index (D) shows a household’s current wealth, but we cannot know whether these assets buffered the household through a shock or were accumulated post-recovery. Section J is different: it explicitly distinguishes strategies used before a shock, actions taken after the shock, and outcomes such as recovery and food sufficiency. This makes J the temporal anchor, while the other sections provide complementary information to explain why some households report more favorable resilience trajectories than others.

The lighter analysis approach focuses on testing associations and describing patterns in the data without making strong causal claims. It uses tools such as cross-tabulations, correlation analysis, and simple regressions to examine how household characteristics, resources, or practices are related to outcomes reported in the resilience section. For example, it can show whether households with more diverse crops also report fewer yield losses, or whether households with access to climate services tend to adopt more pre-shock strategies. This type of analysis is useful for surfacing relationships, identifying potential drivers of resilience, and providing a first layer of evidence. However, because it does not



control for other confounding factors, it can only suggest associations rather than demonstrate that one factor caused another.

The more robust analyses outlined in the table aim to go beyond associations and generate causal insights. This requires controlling for confounding variables, addressing selection bias, and testing the pathways through which interventions influence outcomes. For instance, showing that households with greater assets recover faster demands ruling out that assets are not simply a proxy for other advantages, such as access to better markets or more fertile land. Methods like propensity score matching, instrumental variables, regression discontinuity, mediation analysis, or structural equation modeling can strengthen causal inference, but each comes with assumptions that need to be tested carefully. In practice, robust causal analysis often requires combining survey data with additional data sources (e.g., spatial layers, longitudinal tracking, or natural experiments) and ensuring that confounding influences are explicitly accounted for.

Table A1. Linking survey module in the analysis

Survey section	Links to resilience	Analytical approach (lighter vs more robust)
A. Household identification and characteristics	Provides contextual information on household profile (education, gender, age, farming experience). Not a capacity itself but helps explain heterogeneity in resilience outcomes, such as recovery. future preparedness.	• Cross-tabs or regression of education, gender, or age vs recovery to see whether certain profiles are associated with better outcomes. • Matching or stratified regression to estimate whether household characteristics systematically influence resilience outcomes when confounders are controlled.
B. Household roster	Provides information on household labor pool and skills (absorptive capacity), which shape the ability to respond to shocks and sustain yields or income. Can be linked to yield or income recovery.	• Correlate labor availability with yield or income recovery to test whether larger households bounce back faster. • Instrumental variable regression (e.g., dependency ratio as instrument) to estimate whether labor causally affects recovery.
C. Agricultural land characteristics	Provides information on land size, type, and tenure security that reduce vulnerability and enable adjustments (absorptive/adaptive capacity). Can be tested against recovery or food sufficiency.	• Regression of land size or tenure vs food sufficiency to test whether bigger landholders are less food insecure. • Regression discontinuity or matching to estimate whether tenure security improves recovery.
D. Durable assets	Provides information on financial/physical assets that serve as buffers during shocks (absorptive capacity) and support recovery. Can be tested against income recovery, food access (N), and recovery trajectory.	• Compare recovery outcomes across asset quintiles to see whether wealthier households report faster recovery. • Structural equation modeling to test whether assets improve recovery by enabling adoption of pre- or post-shock strategies.



Survey section	Links to resilience	Analytical approach (lighter vs more robust)
E. Crops grown and sold	Reflects crop diversification and marketing strategies that spread risk and stabilize income (adaptive capacity). Can be tested against yield, income recovery, general recovery, effectiveness or overall food security (N).	• Calculate Shannon diversity index and test associations with yield stability or food security outcomes. • Panel regression or difference-in-differences to estimate whether diversification reduces yield or income losses under shocks.
F. Labour used per crop	Provides information on gendered and hired vs family labor allocation (absorptive/adaptive capacity), which influences workload and recovery. Can be tested against workload changes and recovery.	• Compare gendered labor shares with workload outcomes to see whether women-heavy farms face larger burdens after shocks. • Instrumental variable regression to estimate causal impacts of labor allocation on recovery.
G. Other inputs and amounts used	Provides information on access to fertilizers, pesticides, and manure (adaptive capacity). Input use can buffer yields and income but adds costs and risks. Can be tested against yield recovery and income recovery.	• Regression of input use vs yield or income recovery to see whether higher users report smaller losses. • Propensity score matching to compare adopters vs non-adopters and estimate causal effects on recovery outcomes.
H. Integrated pest management	Shows adoption of pest and disease management practices (adaptive capacity); can be tested against pest losses and yield recovery but also compared to risk reduction to assess whether adopters report higher effectiveness.	• Compare pest damage between IPM adopters and non-adopters to see whether IPM is linked to fewer losses or to perceived effectiveness in risk reduction • Matching or difference-in-differences (if adoption timing is known) to estimate causal effects of IPM.
I. Climate information services and CSA sources	Provides information on access to climate and agricultural advice (anticipatory/adaptive capacity). Explains whether CIS or CSA knowledge enabled households to take pre-shock strategies or post-shock coping actions that improved recovery; can be cross-checked with effectiveness perceptions or future preparedness.	• Test associations between CIS access and adoption of strategies to see whether information users are better prepared; or between CIS access and perceived effectiveness of actions • Mediation analysis to estimate whether CIS access led to practice adoption, which in turn improved recovery outcomes.
K. Other sources of income	Provides information on diversified income sources that buffer agricultural risks (absorptive/adaptive capacity). Can be tested against income stability and food security (N).	• Compare food insecurity between diversified and non-diversified households. • Fixed-effects regression to estimate causal effects of diversification on resilience outcomes.
L. Access to basic services	Provides information on access to markets, finance, insurance, and health services that strengthen preparation and recovery (absorptive/adaptive capacity);	• Logistic regression of service access vs recovery to see whether households near services recover faster. • Natural experiments (e.g., road distance, coverage maps) to estimate



Survey section	Links to resilience	Analytical approach (lighter vs more robust)
	can be tested against recovery and preparedness perceptions.	causal effects of service access on resilience outcomes.
M. Nutrition diversity	Captures household dietary diversity as a direct outcome. Serves as a proxy for food and nutrition security and complements food access and post-shock sufficiency.	• Descriptive HDD scores correlated with shock exposure to see whether diverse diets are associated with resilience. • Mediation analysis to test whether CSA adoption improves dietary diversity, which then improves recovery.
N. Household food access	Captures household food insecurity as a direct outcome. Complements resilience questions (effects post-shock) by showing how households struggled to access food during the year.	• Correlate food insecurity experiences with recovery outcomes to see whether food insecure households recover slower. • Structural equation modeling to test whether assets or diversification affect recovery through food access.
O. General views	Provides information on willingness to adopt innovations (transformative capacity). Signals households' openness to future risk-reducing strategies and long-term resilience orientation.	• Perception analysis by household type to see whether openness to innovations is linked to resilience perceptions. • Discrete choice models to estimate whether preferences for drought-resistant seeds or CSA predict adoption and future resilience.

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